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RESEARCH ARTICLE

An Improved Ensemble Method With Data Resampling for Credit Risk Prediction

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ABSTRACT The increasing complexity and dynamic nature of financial data present significant challenges in accurately predicting credit risk, a critical task in the banking and finance sector. The application of machine learning (ML) in credit risk prediction has been hindered by the imbalanced nature of credit datasets. This study proposes an improved approach for predicting credit risk using a stacked ensemble method combined with a hybrid data resampling technique. The ensemble comprises random forests, logistic regression, and a convolutional neural network (CNN) as base learners, with the multilayer perceptron (MLP) serving as a meta-learner. To address the data imbalance, the Synthetic Minority Over-sampling Technique and Edited Nearest Neighbors (SMOTE-ENN) technique were applied. The proposed approach is benchmarked against other well-performing classifiers, including random forest, logistic regression, MLP, and CNN. The integration of hybrid data resampling with a robust stacking ensemble significantly enhanced credit risk prediction, with the proposed approach achieving sensitivity and specificity of 0.921 and 0.946 for the Australian dataset and 0.928 and 0.891 for the German dataset. Also, the stacked classifier achieved a sensitivity and specificity of 0.000 and 1.000 before data resampling for the Credit Risk Classification dataset with an accuracy of 0.7644. After data resampling, the accuracy, sensitivity, and specificity are 0.8056, 0.7989 and 0.8125, respectively. On the other hand, using the credit risk analysis for the extended banking loans dataset, the accuracy, sensitivity and specificity of the stacked classifier before data resampling are 0.8429, 0.6316, and 0.9216, respectively. After data resampling, the accuracy, sensitivity and specificity scores of the stacked classifier trained using the credit risk analysis for the extended banking loans dataset are 0.9632, 1.0000, and 0.9242, respectively. This shows that after data resampling, the performance of the stacked classifier trained using the credit risk analysis for the extended banking loans dataset outperformed other models.

INDEX TERMS Credit risk, CNN, ensemble learning, machine learning, MLP, random forest.

I. INTRODUCTION

Credit risk prediction is a critical challenge in the financial sector, with significant implications for banks and lending institutions globally [1], [2], [3]. It is estimated that poor credit risk assessment can lead to substantial financial losses, impacting the stability and profitability of financial institutions. The ability to accurately predict which applicants are likely to default on their loans is essential for managing risk and ensuring the long-term sustainability of credit systems [4], [5], [6]. The dynamic and complex nature of credit risk, influenced by a multitude of factors, including economic conditions and individual financial behaviours, necessitates the adoption of more advanced and adaptive strategies beyond traditional risk assessment methods.

The integration of machine learning (ML) techniques in credit risk prediction has revolutionized the approach to data analysis and decision-making [7], [8]. ML offers substantial improvements over conventional statistical methods by enabling the analysis of vast amounts of data to identify patterns that may indicate creditworthiness or potential default. Machine learning algorithms, ranging from

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logistic regression to complex neural networks, have the capability to learn from historical data and make informed predictions about new, unseen applicants, thereby enhancing the accuracy and efficiency of credit risk assessment systems.

However, one of the significant challenges in applying ML to credit risk prediction is the imbalanced nature of the datasets [9], [10], [11]. Typically, the number of applicants who default is much smaller compared to those who do not, leading to models that are biased towards predicting the majority class and failing to identify a substantial number of potential defaults [12]. Addressing this issue requires sophisticated data resampling techniques and advanced ensemble methods that can effectively balance the dataset and improve the model's sensitivity towards the minority class.

This study proposes an improved ensemble method that combines the strengths of multiple ML techniques, namely random forests, logistic regression, and a convolutional neural network (CNN), with a multilayer perceptron (MLP) serving as the meta-learner. This stacked ensemble approach is designed to leverage the diverse strengths of each base learner, enhancing the model's ability to capture and analyze complex patterns in credit risk data. Additionally, the Synthetic Minority Over-sampling Technique - Edited Nearest Neighbors (SMOTE-ENN), a hybrid resampling approach, is employed to handle the class imbalance, ensuring that the model is not biased toward the majority class.

Furthermore, the selection of base learners in the stacked ensemble is strategically designed to harness diverse analytical strengths. For instance, random forests are known for their robustness and ability to handle non-linear relationships, making them highly effective in classification tasks [13]. Logistic regression, despite its simplicity, offers fast and interpretable results, making it ideal for initial insights and baseline performance metrics [14]. Meanwhile, CNNs excel in identifying complex patterns and correlations in large and high-dimensional datasets, which can be creatively applied to feature-rich financial data [15], [16]. The multilayer perceptron, used as the meta-learner, integrates these diverse inputs into a coherent output, refining the final decision process through its layered structure and nonlinear processing capabilities.

The anticipated outcome of this research is not only to advance the state of the art in credit risk prediction but also to provide a framework that can be adapted to other domains facing similar challenges with imbalanced data. By demonstrating the effectiveness of hybrid data resampling techniques combined with a stacked ensemble approach, this study aims to encourage broader adoption and further exploration in both academic and industry settings. Such advancements are crucial in the ongoing effort to improve credit risk assessment, ultimately leading to more efficient resource utilization and greater stability in the financial sector.

The research questions to be addressed in this study are:

1) Does a stacked ensemble model improve the accuracy and reliability of credit risk prediction compared to individual machine learning models?
2) What is the impact of using SMOTE-ENN in handling class imbalance in credit risk prediction datasets?
3) How does the proposed approach compare to existing state-of-the-art models in terms of classification performance metrics?

The rest of the paper is structured as follows: Section II discusses related works, while Section III presents the datasets and ML methods used in this study. Section IV presents the proposed credit risk prediction approach, while Section V highlights the results and discussion, and Section VI concludes the study.

II. RELATED WORKS

Credit risk models have been a core of risk management in the financial industry, with traditional models predominantly relying on statistical methods such as logistic regression and decision trees [17], [18], [19]. These methods, while simple and interpretable, often struggle with capturing the complex, non-linear relationships inherent in financial data, particularly as datasets grow in size and complexity [20], [21]. The limitations of these classical approaches have prompted a shift towards more advanced machine learning (ML) techniques.

With the advent of big data and increased computational power, ensemble methods such as random forests, gradient boosting machines, and extreme gradient boosting (XGBoost) have gained prominence in credit risk prediction due to their ability to handle large datasets and complex relationships more effectively [22], [23], [24]. However, these methods are not without challenges. They often require extensive data preprocessing and parameter tuning to avoid overfitting, and their interpretability is often limited compared to traditional models.

Deep learning, an advanced subset of ML, has been increasingly applied to credit risk prediction due to its ability to model intricate patterns within large datasets [25]. Techniques such as CNNs and Recurrent Neural Networks (RNNs) have shown significant promise in processing sequential and spatial data, thereby enhancing predictive performance [26]. Despite their success, deep learning models are frequently criticized for their "black-box" nature, which can complicate their use in regulated financial environments where model interpretability is crucial [27], [28], [29].

Another critical challenge in credit risk prediction is the imbalance in datasets, where the number of default cases is significantly lower than non-default cases. This imbalance can lead to biased models that favor the majority class, thereby reducing the model's ability to accurately predict defaults [30]. Traditional resampling methods, such as oversampling and undersampling, have been used to address this issue. However, these techniques can lead to overfitting or loss of valuable information, respectively [31], [32], [33].

To address the imbalance issue more effectively, recent studies have explored hybrid resampling techniques like the Synthetic Minority Over-sampling Technique combined with Edited Nearest Neighbors (SMOTE-ENN) [34]. SMOTE-ENN not only generates synthetic instances to balance the minority class but also removes potentially noisy data points that could negatively impact model performance. This approach has been shown to improve the sensitivity of ML models to the minority class without sacrificing overall model accuracy.

While significant advancements have been made in the application of ML in credit risk prediction, challenges remain, particularly in dealing with highly imbalanced datasets and reaching optimal performance. The proposed stacked ensemble method in this study seeks to address these challenges by employing advanced resampling and leveraging the strengths of individual models (such as CNN, random forest, and logistic regression) to build an ensemble model with enhanced performance.

III. MATERIALS AND METHODS

A. DATASETS

This study employs two well-known datasets for evaluating the performance of the machine learning models: the Australian and the German credit datasets. The Australian Credit Dataset is a widely used benchmark in credit risk modelling [35]. It consists of 14 features that describe various aspects of a loan applicant's financial status and personal information, along with a binary target variable that indicates credit risk. The dataset includes both categorical and numeric data types, covering aspects such as credit history, savings, employment status, and loan details. The target variable, labelled "Class," identifies whether a credit applicant is classified as good or bad risk, providing a foundation for testing classification algorithms. The dataset is described in Table 1.

TABLE 1. Australian credit dataset features.

Feature	Description	Data Type
A1	Status of existing checking account	Categorical
A2	Duration of Loan in months	Numeric
A3	Credit history	Categorical
A4	Purpose of the Loan	Categorical
A5	Loan amount	Numeric
A6	Savings account	Categorical
A7	Length of employment	Categorical
A8	Installment rate	Numeric
A9	Sex	Categorical
A10	Other Loans	Categorical
A11	Present residence duration	Numeric
A12	Property	Categorical
A13	Age (years)	Numeric
A14	Other installment plans	Categorical
Class	Credit risk (1 = good, 2 = bad)	Binary

The German Credit Dataset is a critical resource for evaluating credit risk prediction models. It comprises 10 features that capture key financial and personal information about loan applicants, alongside a binary target variable that classifies credit risk [36]. The dataset includes both categorical and numeric variables, such as account status, credit amount, employment duration, and housing conditions. The target variable, "Credit Risk," differentiates between good and bad credit applicants, making this dataset a valuable benchmark for testing and comparing classification algorithms in the field of credit scoring. It is tabulated in (Table 2).

The Credit Risk Analysis for extended Banking Loans dataset is critical for training and evaluating the credit risk prediction models [37]. The dataset comprises of 9 features which are all numeric. These features include age, ed (education level), employ (work experience), address (address of the customer), income (yearly income of the customer), debtinc (debt of income ratio), creddebt (Credit to Debt ratio), othdebt (Other debts), default (Customer defaulted in the past (1 = defaulted, 0 = Never defaulted)). It is tabulated in (Table 3).

The Credit Risk Classification dataset is a customer transaction and demographic-related data which holds risky and non-risky customer information for specific banks [38]. It has both numerical and categorical features, but was encoded. The features include: label, id, fea_1, fea_2, fea_3, fea_4, fea_5, fea_6, fea_7, fea_8, fea_9, fea_10, fea_11. The fea_1 to fea_11 are the encoded values of the features, while "label" is the target variable. The features of the dataset, as well as its descriptions and data type, are tabulated in (Table 4)

B. MACHINE LEARNING CLASSIFIERS

The algorithms used in this study include random forest, CNN, and logistic regression, and they are discussed below.

1) RANDOM FOREST

Random Forest is a robust ensemble learning method that operates by constructing multiple decision trees during the training phase and outputting the class that is the mode of the classifications suggested by the individual trees [39]. This methodology is encapsulated in the equation:

$$\hat{y} = \text{mode}(\{h(x, \Theta_k)\}_{k=1}^{K}) \tag{1}$$

where $\hat{y}$ is the predicted outcome, $h(x, \Theta_k)$ denotes the prediction of the k-th tree, and K represents the total number of trees [40]. Random forest performs both classification and regression with high accuracy. It is particularly effective due to its ability to reduce overfitting by averaging multiple trees, handle large datasets with higher dimensionality, and maintain accuracy even when a significant proportion of the data is missing [41], [42]. In the ensemble context, Random Forest provides a diversified model response, crucial for capturing a wide spectrum of fraud indicators in imbalanced datasets.

2) CONVOLUTIONAL NEURAL NETWORK

CNNs are specifically tailored for parsing through data with a grid-like topology, such as images [43]. CNNs utilize

TABLE 2. German credit dataset features.

Feature	Description	Data Type
Checking account	Status of the applicant's checking account	Categorical
Duration	Length of the credit in months	Numeric
Purpose	Reason for the credit	Categorical
Loan amount	Loan amount requested	Numeric
Savings account	Status of the customer's savings account	Categorical
Age	Age of the customer (years)	Numeric
Employment	Duration of the applicant's employment	Categorical
Sex	Gender of the customer	Categorical
Housing	customer's housing status	Categorical
Job	Category of the customer's job	Categorical
Credit Risk	Class variable indicating credit risk (1=good, 2=bad)	Binary

TABLE 3. The credit risk analysis for extended banking loans dataset.

Feature	Description	Data Type
Age	Age of the customer	Numeric
Ed	Education level of the customer	Numeric
Employ	Work experience of the customer	Numeric
Address	Address duration of the customer	Numeric
Income	Yearly income of the customer	Numeric
DebtInc	Debt-to-income ratio	Numeric
CredDebt	Credit-to-debt ratio	Numeric
OthDebt	Other debts of the customer	Numeric
Default	Customer default history (1 = Defaulted, 0 = Never Defaulted)	Binary

TABLE 4. The credit risk classification dataset.

Feature	Description	Data Type
Fea_1	Categorical feature 1	Categorical
Fea_3	Categorical feature 3	Categorical
Fea_5	Categorical feature 5	Categorical
Fea_6	Categorical feature 6	Categorical
Fea_7	Categorical feature 7	Categorical
Fea_9	Categorical feature 9	Categorical
Label	Customer credit risk (1 = High, 0 = Low)	Binary

multiple layers of convolutional operations that filter inputs for useful information, transforming them through a hierarchical structure of layers to capture and recombine properties into higher-level features [44], [45], [46]. These operations are typically followed by non-linear activation functions and pooling to reduce spatial size. The convolutional layer equation can be described as:

$$f(x) = \max(0, x * W + b) \tag{2}$$

where x is the input matrix, W is the kernel or filter matrix, and b is the bias. CNNs are excellent for feature extraction and capable of detecting intricate patterns without the need for manual feature engineering, making them particularly advantageous for complex datasets like those in healthcare [47], [48]. Within the proposed ensemble, a CNN can extract subtle pattern-based discrepancies indicative of fraudulent activities, which might be missed by other models.

3) LOGISTIC REGRESSION

Logistic regression is a predictive analysis algorithm used primarily for binary classification [49]. It models the probability of a default class (usually the "positive" class) based on input features using the logistic function or sigmoid:

$$p(y = 1) = \frac{1}{1 + e^{-(\beta_0+\beta_1 x)}} \tag{3}$$

where β_0 and β_1 are coefficients for the intercept and the input features, respectively [50]. Logistic Regression is valued for its simplicity and effectiveness in providing probability scores regarding class membership. It is especially useful in an ensemble as it provides a linear decision boundary and can serve as a calibration model, giving interpretable results that can be critical in understanding how input variables relate to the probability of fraud.

4) MULTILAYER PERCEPTRON

The MLP is a fundamental type of neural network consisting of at least three layers: an input layer, one or more hidden layers, and an output layer [51]. Each neuron in these layers applies a nonlinear activation function, typically ReLU or sigmoid, to the weighted sum of its input:

$$a = \sigma(Wx + b) \tag{4}$$

where σ represents the activation function, W is the weight matrix associated with inputs x, and b is the bias vector. MLPs are capable of learning complex nonlinear hypotheses over the input feature space [52], [53]. As a meta-learner in a stacked ensemble, the MLP integrates diverse predictions from the base models, effectively capturing interactions and dependencies that might be too complex for any single model to learn, thereby significantly enhancing the ensemble's predictive power and robustness.

IV. PROPOSED CREDIT RISK PREDICTION METHOD

A. DATA RESAMPLING

In credit risk prediction, the prevalence of imbalanced datasets, where default cases are much rarer than non-default ones, presents significant challenges. Such imbalances can lead to models that are biased towards predicting the majority class, reducing the effectiveness of risk assessment. To address this imbalance, we utilize the SMOTE-ENN method as described in Algorithm 1. In the SMOTE-ENN implementation, SMOTE is employed to artificially augment the minority class by generating synthetic examples rather than by simple replication. This oversampling technique selects samples from the minority class and generates new instances that are similar but not identical, thus enriching the dataset without copying exact duplicates. This method aids in expanding the decision region of the minority class, making it more general. Meanwhile, ENN is used to refine the dataset by removing instances of the majority class that are likely to be noise or are close to the decision boundary. This undersampling technique enhances the quality of the dataset by ensuring that only the most representative instances of the majority class are retained, which helps in defining clearer class boundaries. The SMOTE-ENN technique aims to effectively balance the dataset by extending the minority class data points and pruning overlapping majority class points. This dual approach not only mitigates the issues associated with class imbalance but also improves the model's ability to predict credit risk by making the classes more distinct. The SMOTE-ENN method is particularly vital for our stacked ensemble model as it enhances the dataset's quality, ensuring that the base learners in the ensemble have a more balanced and noise-free dataset to train on, thereby improving the sensitivity and specificity of the credit risk prediction system.

B. STACKED ENSEMBLE

The proposed approach to credit risk prediction, described in Algorithm 2, employs a stacked ensemble method designed to leverage the unique strengths of diverse ML algorithms to enhance predictive accuracy. The ensemble framework is particularly advantageous in complex credit risk prediction scenarios where different types of default risk may be detected more effectively by different algorithms. The base learners in this ensemble, specifically chosen for their complementary capabilities, include random forest, logistic regression, and a CNN. The random forest is integral to the ensemble due to its high accuracy and robustness. It is effective in handling large datasets with numerous features, and it provides important measures of feature importance, thus offering insights into the most discriminative predictors of credit risk.

Meanwhile, logistic regression offers a fast and robust benchmark for any classification problem. The logistic model also serves to linearly separate default from non-default cases, contributing to the ensemble by providing a different perspective compared to more complex models. On the other hand, CNN can identify intricate patterns and anomalies in data, which might be indicative of underlying risk factors that are difficult to detect with more traditional machine learning approaches. The outputs of these diverse base learners are then fed into a Multilayer Perceptron (MLP), which acts as the meta-learner. The MLP is crucial for integrating insights gained from each of the base models. It processes the predictions (meta-features) from the initial models to produce a final prediction, effectively capturing complex interactions between the predictions that might be missed by any single model. This layered approach allows the ensemble to benefit from the individual strengths of each base learner.

Algorithm 1 SMOTE-ENN Method

Input: Raw credit records dataset D.
Output: Balanced credit records dataset.
Steps
1. Oversampling (SMOTE)
1) Select a random sample x_i from the minority class.
2) Find the K nearest neighbours of x_i.
3) For each neighbour q, generate a synthetic data instance p as follows:
$$p = x_i + \lambda(q - x_i)$$
where λ is a random number between 0 and 1.
4) Assign the minority class label to p.
5) Continue generating synthetic samples until the desired class balance is achieved.

2. Undersampling (ENN)
6) For each instance x_r in the dataset D, determine the k-nearest neighbours (with $k = 3$).
7) If the majority of x_r's neighbours are from a different class, remove x_r from D.
8) Repeat for all instances in D to reduce noise and remove misclassified samples.

This stacked approach not only improves model accuracy but also enhances the ability to generalize across different types of credit risks, thus significantly reducing the likelihood of model overfitting to peculiarities in the training data. By integrating multiple models, the ensemble is better equipped to handle the complexities and variations inherent in credit risk prediction, resulting in more robust and reliable predictions.

To ensure reproducibility, the experimental setup was carefully structured, incorporating key processes such as feature selection, data resampling, cross-validation, and the use of appropriate software tools. Standardization was performed using `StandardScaler()`, which ensured that each feature had a mean of zero and unit variance, thereby optimizing

Algorithm 2 Stacked Ensemble Method

Input: Training data $\mathcal{D}_{train}$ and testing data $\mathcal{D}_{test}$
Output: Final predictions $\mathcal{P}$
Step 1: Train Base Learners
Train the following models on $\mathcal{D}_{train}$:
1) Random Forest
2) Logistic Regression
3) Convolutional Neural Network (CNN)
Step 2: Generate Meta-features
for each instance x_i in $\mathcal{D}_{train}$ **do**
Predict $p_{RF}(x_i)$, $p_{LR}(x_i)$, and $p_{CNN}(x_i)$ using the trained models
Form a new feature vector $z_i = [p_{RF}(x_i), p_{LR}(x_i), p_{CNN}(x_i)]$
end for
Step 3: Train Meta-learner
Train a Multilayer Perceptron (MLP) on the new dataset $\{z_i, y_i\}$ where y_i are the true labels.
Step 4: Prediction on Test Data
for each instance x_j in $\mathcal{D}_{test}$ **do**
Predict $p_{RF}(x_j)$, $p_{LR}(x_j)$, and $p_{CNN}(x_j)$
Form a new feature topic $z_j = [p_{RF}(x_j), p_{LR}(x_j), p_{CNN}(x_j)]$
Predict final output $\hat{y}_j$ using the trained MLP
Append $\hat{y}_j$ to $\mathcal{P}$
end for

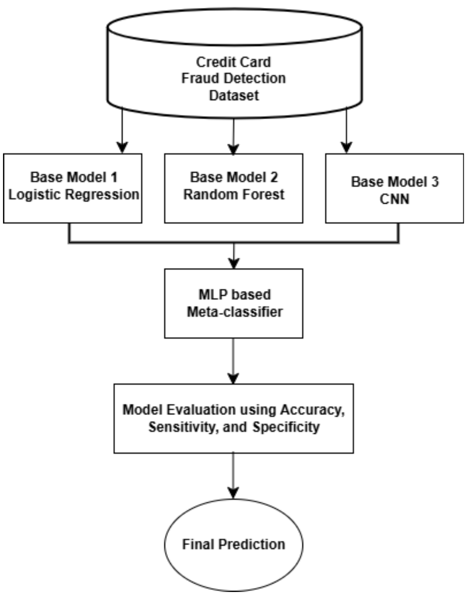

FIGURE 1. Structure of the proposed ensemble.

model performance. Given the presence of class imbalance, SMOTEENN was employed as a resampling strategy. This technique combined Synthetic Minority Over-sampling Technique (SMOTE), which generated synthetic samples for the minority class using `k_neighbors=5`, with Edited Nearest Neighbors (ENN), which removed ambiguous samples to enhance dataset quality. Following resampling, the dataset was split into 80% training and 20% testing using `train_test_split(test_size=0.2, random_state=42)`, ensuring consistency in experimental results.

Although explicit k-fold cross-validation was not applied, the train-test split provided an effective means of assessing model generalization. The deep learning models implemented in this study were a Convolutional Neural Network (CNN) and a Multi-Layer Perceptron (MLP) model, which featured a 1D convolutional layer with 64 filters and a kernel size of 3, followed by a flattening operation and fully connected dense layers. The final output layer employed the softmax activation function, enabling multi-class classification. The model was trained using the Adam optimizer, with `categorical_crossentropy` as the loss function, a batch size of 16, and a total of 50 epochs, ensuring a balance between computational efficiency and model performance. The entire experimental workflow was executed using Python 3.12, with essential libraries such as NumPy and Pandas for data manipulation, scikit-learn for preprocessing and model evaluation, imbalanced-learn for resampling, and TensorFlow/Keras for deep learning model implementation.

EXPERIMENTAL SETUP AND HYPERPARAMETERS

To ensure reproducibility, the experimental setup was carefully structured, incorporating key processes, such as feature selection, data resampling, cross-validation, and the use of appropriate software tools. Standardization was performed using `StandardScaler()`, which ensured that each feature had a mean of zero and unit variance, thereby optimizing model performance. Given the presence of class imbalance, SMOTEENN was employed as a resampling strategy.

TABLE 5. Hyperparameters used in the experiment.

Hyperparameter	Value
Standardization	`StandardScaler()`
SMOTE k-neighbors	5
Train-test split ratio	80%-20%
Random state	42
CNN filters	64
Kernel size	3
Batch size	16
Epochs	50
Loss function	`categorical_crossentropy`
Optimizer	Adam

V. RESULTS AND DISCUSSION

This section presents the results for the various models. The models were evaluated before and after feature selection using the Australian dataset, German dataset, Credit Risk Classification Dataset, and Credit Risk Analysis for the extended Banking Loans dataset. The following performance evaluation metrics will be used in the course of this study: accuracy, sensitivity and specificity.

A. PERFORMANCE OF THE MODELS USING THE AUSTRALIAN DATASET

The performance of the models using the Australian dataset, presented in Tables 6 and 7, clearly illustrates the substantial improvements that resampling techniques bring to model accuracy, recall, and specificity. Before resampling, the proposed stacked ensemble model already outperformed the other models with an accuracy of 0.906, a recall of 0.891, and a specificity of 0.914. In contrast, other models, such as CNN, random forest, MLP, and logistic regression, displayed lower performance, especially in balancing sensitivity and specificity, indicating challenges in effectively managing the class imbalance inherent in the dataset.

TABLE 6. Model performance before data Resampling using australian dataset.

Model	Accuracy	Recall	Specificity
CNN	0.860	0.855	0.880
Random forest	0.865	0.850	0.889
MLP	0.820	0.848	0.779
Logistic Regression	0.846	0.830	0.853
Proposed Stacked Ensemble	0.906	0.891	0.914

TABLE 7. Model performance after data Resampling using australian dataset.

Model	Accuracy	Recall	Specificity
CNN	0.881	0.905	0.910
Random forest	0.914	0.890	0.926
MLP	0.876	0.899	0.864
Logistic Regression	0.883	0.904	0.917
Proposed Stacked Ensemble	0.930	0.921	0.946

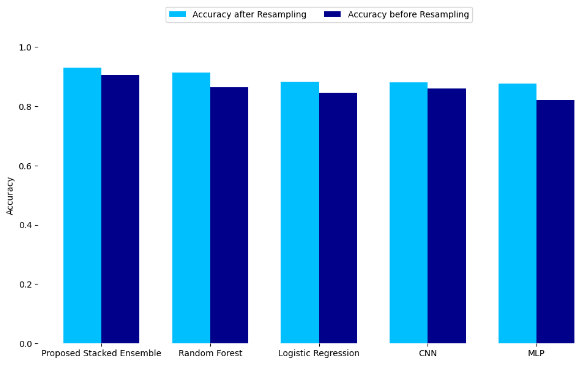

FIGURE 2. Accuracy comparison for australian dataset.

Figure 2 compares the accuracy of five models on the Australian dataset before and after resampling. The proposed stacked ensemble model had the highest accuracy, which outperformed other models. Figure 3 compares the recall of five models on the Australian dataset before and after resampling. The proposed stacked ensemble model had the highest recall, which outperformed other models. Figure 4 compares the specificity of five models on the Australian dataset before and after resampling. The proposed stacked ensemble model had the highest specificity, which outperformed other models.

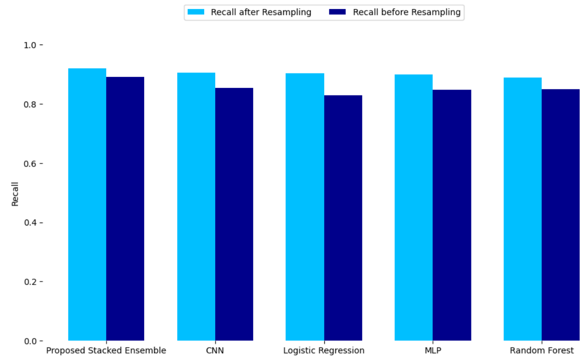

FIGURE 3. Recall comparison for australian dataset.

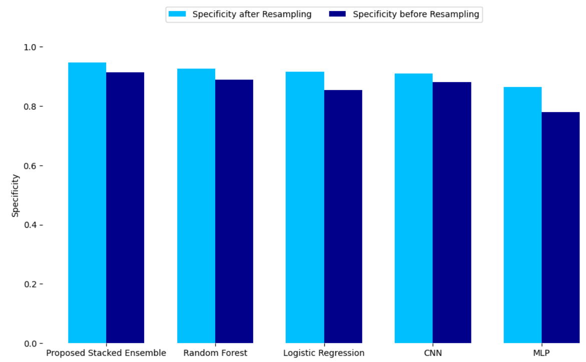

FIGURE 4. Specificity comparison for australian dataset.

Figure 5 compares the accuracy of five models on the German dataset before and after resampling. The proposed stacked ensemble model had the highest accuracy, which outperformed other models. Figure 6 compares the recall of five models on the German dataset before and after resampling. The proposed stacked ensemble model had the highest recall, which outperformed other models. Figure 7 compares the specificity of five models on the German dataset before and after resampling. The proposed stacked ensemble model had the highest specificity, which outperformed other models.

Post-resampling, all models experienced marked performance enhancements (as shown in Figures 2 - 4), with the proposed stacked ensemble achieving the highest metrics: 0.930 in accuracy, 0.921 in recall, and 0.946 in specificity. These results demonstrate the ensemble's ability to effectively leverage the balanced dataset, providing superior generalization and robustness compared to individual models. The ensemble's integration of diverse base learners allows it to capture complex patterns in the data, leading to more accurate and reliable credit risk predictions. This demonstrates the critical role of ensemble methods in enhancing model performance, particularly in domains like credit risk assessment, where class imbalance is a significant issue.

B. PERFORMANCE OF THE MODELS USING THE GERMAN DATASET

The performance of the models using the German dataset, as illustrated in Tables 8 and 9, indicates the significant impact of resampling techniques on improving model accuracy, recall, and specificity. Before resampling, the proposed stacked ensemble model outperformed all other models, achieving an accuracy of 0.811, a recall of 0.799, and a specificity of 0.486. However, other models, such as CNN, random forest, MLP, and logistic regression, exhibited lower performance metrics, particularly in specificity, indicating difficulties in correctly identifying negative cases in the imbalanced dataset.

TABLE 8. Model performance before data Resampling using german.

Model	Accuracy	Recall	Specificity
CNN	0.749	0.779	0.402
Random forest	0.728	0.765	0.417
MLP	0.710	0.784	0.398
Logistic Regression	0.755	0.747	0.440
Proposed Stacked Ensemble	0.811	0.799	0.486

TABLE 9. Model performance after data Resampling using german.

Model	Accuracy	Recall	Specificity
CNN	0.790	0.830	0.711
Random forest	0.823	0.860	0.734
MLP	0.825	0.829	0.807
Logistic Regression	0.811	0.825	0.729
Proposed Stacked Ensemble	0.914	0.928	0.891

After applying resampling methods, all models demonstrated substantial improvements as shown in Figures 5 - 7, with the proposed stacked ensemble model achieving the highest accuracy (0.914), recall (0.928), and specificity (0.891). This enhancement is particularly significant when compared to the CNN and random forest models, which, despite improvements, did not match the performance of the stacked ensemble. The results highlight the superior ability of the stacked ensemble model to generalize from resampled data, effectively addressing the class imbalance problem. The ensemble's combination of diverse base models enables it to capture various aspects of the data, leading to more reliable predictions across all metrics. This performance demonstrates the robustness and effectiveness of the proposed approach in credit risk prediction, particularly when dealing with imbalanced datasets.

C. PERFORMANCE OF THE MODELS USING THE CREDIT RISK CLASSIFICATION DATASET

The performance of the models using the Credit Risk Classification dataset is presented in Table 10 and Table 11.

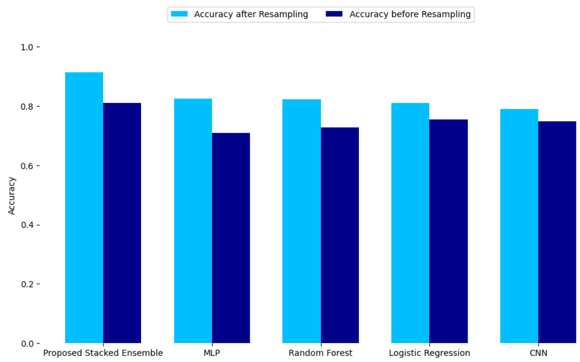

FIGURE 5. Accuracy comparison for german dataset.

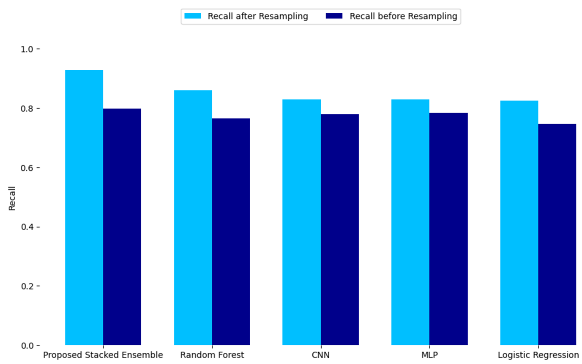

FIGURE 6. Recall comparison for german dataset.

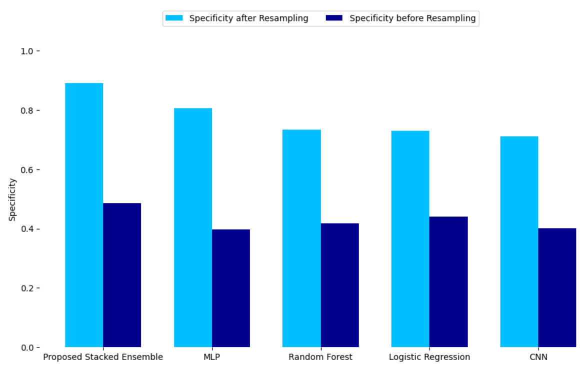

FIGURE 7. Specificity comparison for german dataset.

Table 10 presents the model performance before data resampling using the credit risk classification dataset. Table 11 presents the model performance after data resampling using the credit risk classification dataset. Figure 11 shows the ROC curves for each model performance after resampling. Before resampling, all models, including the CNN, MLP, Logistic Regression, and the Proposed Stacked Ensemble, achieved similar accuracy (around 76.44). However, they exhibited poor recall (0.0000-0.0377), indicating that they failed to correctly identify default cases. This suggests a strong class imbalance, where models predominantly classified instances as non-default, achieving high specificity (close to 1.0000) but failing to detect defaulters. After resampling, there was a significant improvement in recall across all models, with the Proposed Stacked Ensemble (0.7989) and Random Forest (0.8098) performing the best in detecting defaults. While accuracy slightly decreased for some models due to more balanced predictions, specificity remained relatively high, particularly for the Proposed Stacked Ensemble (0.8125) and Random Forest (0.8125).

TABLE 10. Model performance before data Resampling using credit risk classification dataset.

Model	Accuracy	Recall	Specificity
CNN	0.7644	0.0000	1.0000
Random Forest	0.7556	0.0377	0.9767
MLP	0.7644	0.0000	1.0000
Logistic Regression	0.7644	0.0000	1.0000
Proposed Stacked Ensemble	0.7644	0.0000	1.0000

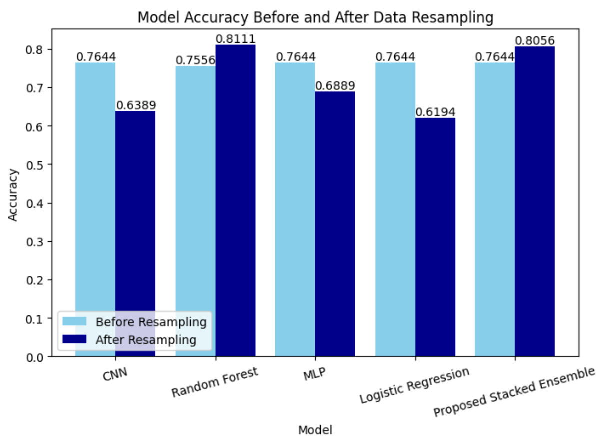

FIGURE 8. **Accuracy comparison for credit risk classification dataset.**

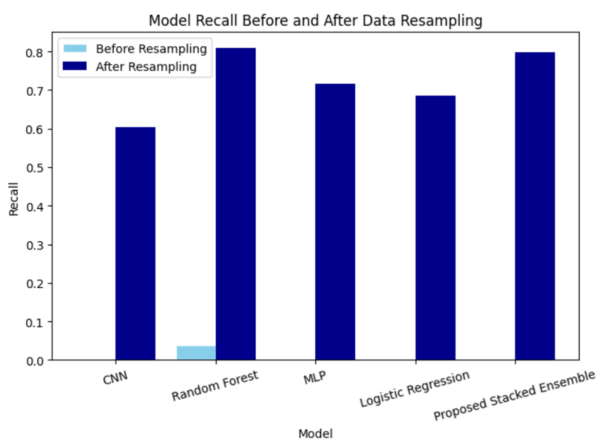

FIGURE 9. **Recall comparison for credit risk classification dataset.**

D. PERFORMANCE OF THE MODELS USING THE CREDIT RISK ANALYSIS FOR EXTENDED BANK LOANS DATASET

The results in Table 14 indicate the model performance before data resampling, showing relatively lower recall scores

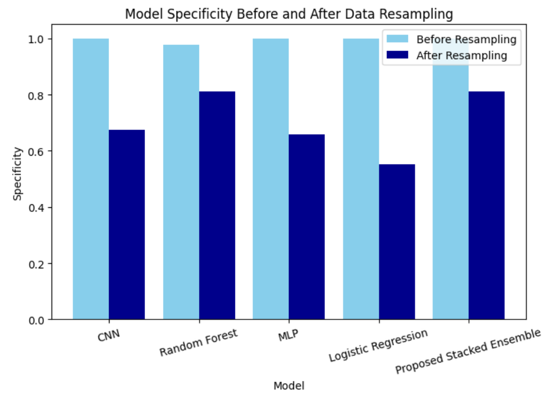

FIGURE 10. **Specificity comparison for credit risk classification dataset.**

TABLE 11. Model performance after data Resampling using credit risk classification dataset.

Model	Accuracy	Recall	Specificity
CNN	0.6389	0.6033	0.6761
Random Forest	0.8111	0.8098	0.8125
MLP	0.6889	0.7174	0.6591
Logistic Regression	0.6194	0.6848	0.5511
Proposed Stacked Ensemble	0.8056	0.7989	0.8125

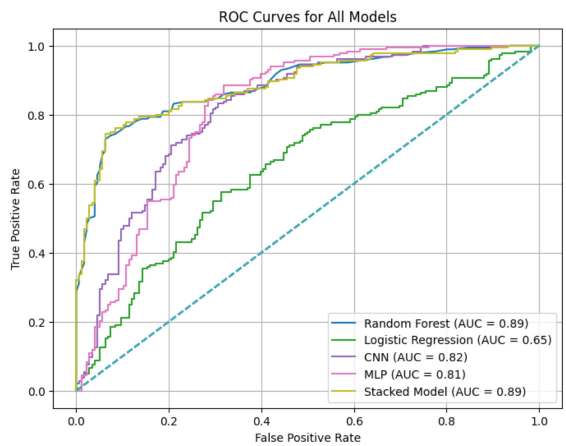

FIGURE 11. **ROC curves for credit risk classification dataset.**

across all models, which suggests difficulty in correctly identifying high-risk cases, while Figure 15 shows the ROC curves for each models after resampling. The proposed stacked ensemble model achieved the highest recall (0.6316), demonstrating better sensitivity than other models. Table 13 presents the performance after data resampling, revealing a significant improvement in recall, with most models achieving 1.0000, indicating perfect detection of high-risk cases. Accuracy and specificity also increased, particularly for the proposed stacked ensemble model, which attained the highest accuracy (0.9632) and specificity (0.9242). This highlights the effectiveness of data resampling in improving model robustness for credit risk analysis in extended banking loans.

TABLE 12. Model performance before data Resampling using credit risk analysis for extended banking loans dataset.

Model	Accuracy	Recall	Specificity
CNN	0.8500	0.5526	0.9608
Random Forest	0.7786	0.3947	0.9216
MLP	0.8071	0.5263	0.9118
Logistic Regression	0.8571	0.5789	0.9608
Proposed Stacked Ensemble	0.8429	0.6316	0.9216

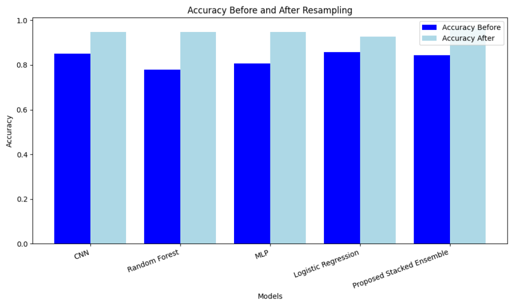

FIGURE 12. Accuracy comparison for credit risk analysis for extended banking loans dataset.

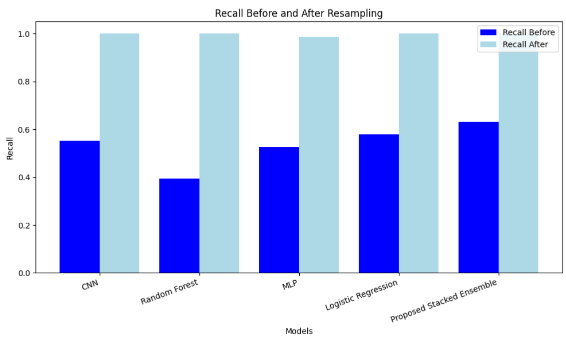

FIGURE 13. Recall comparison for credit risk credit risk analysis for extended banking loans dataset.

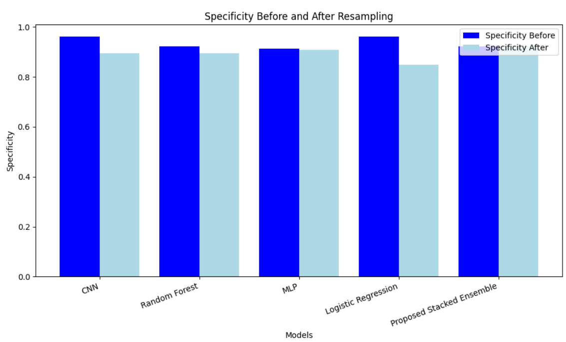

FIGURE 14. Specificity comparison for credit risk credit risk analysis for extended banking loans dataset.

E. COMPARISON WITH RECENT LITERATURE

The performance of the models using the Australian and German datasets Tables 14 and 15 The proposed stacked ensemble method demonstrates superior performance compared to recent studies on both the Australian and German credit datasets. Specifically, the stacked ensemble achieves an accuracy of 0.930 on the Australian dataset and 0.914 on the German dataset, surpassing all other models in the comparison. This method also shows enhanced sensitivity and specificity, indicating its robustness in distinguishing between default and non-default cases. The combination of diverse base learners—random forest, logistic regression, and CNN—enables the model to capture various patterns in the data, leading to a more comprehensive understanding of credit risk.

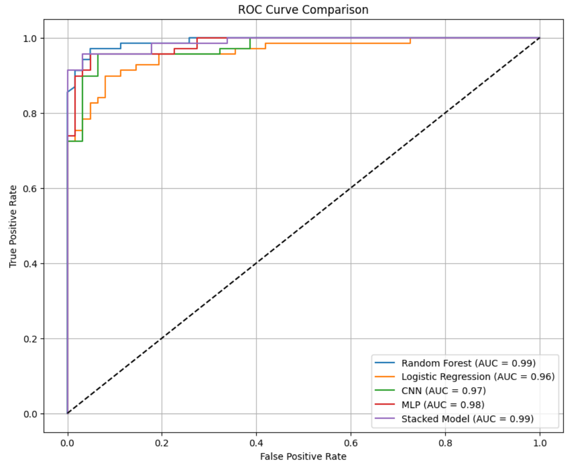

FIGURE 15. ROC curves for credit risk analysis for extended bank loans dataset.

TABLE 13. Model performance after data Resampling using credit risk analysis for extended banking loans dataset.

Model	Accuracy	Recall	Specificity
CNN	0.9485	1.0000	0.8939
Random Forest	0.9485	1.0000	0.8939
MLP	0.9485	0.9857	0.9091
Logistic Regression	0.9265	1.0000	0.8485
Proposed Stacked Ensemble	0.9632	1.0000	0.9242

The stacked ensemble outperforms other methods primarily due to its ability to integrate the strengths of multiple models. Each base learner contributes unique insights, with random forests handling feature-rich datasets effectively, logistic regression providing a linear perspective, and CNNs detecting complex patterns. The meta-learner (MLP) further refines these predictions by combining the outputs of the base models, resulting in a more accurate and reliable final prediction. This multi-layered approach not only enhances model accuracy but also improves the generalizability of the predictions, making the stacked ensemble particularly effective in handling the complexities of credit risk assessment.

F. DISCUSSION

The results from the previous sections show the substantial improvements in model performance achieved through the

TABLE 14. Comparison with recent studies using australian dataset.

Reference	Method	Accuracy	Sensitivity	Specificity
[54]	Evolutionary Extreme Learning Machine	0.871	0.859	0.878
[55]	Boosted LR Ensemble	0.880	-	-
[56]	DT and SMOTE	0.838	-	-
[56]	DT and ADASYN	0.816	-	-
[57]	XGBoost with Grid Search	0.890	-	-
[58]	Gaussian Mixture Classifier	0.867	0.898	-
[59]	Rotation forest	0.865	-	-
[60]	MLP and Feature selection	0.853	0.854	0.858
[59]	Random Subspace DT	-	0.869	-
[61]	CNN	0.880	-	-
[62]	Bagging Ensemble	0.865	-	-
This study	Proposed Stacked Ensemble	0.930	0.921	0.946

TABLE 15. Comparison with recent studies using german dataset.

Reference	Method	Accuracy	Sensitivity	Specificity
[58]	Gaussian Mixture Classifier	0.742	0.756	-
[63]	LSTM Ensemble with resampling	0.904	-	-
[57]	XGBoost with Grid Search	0.816	-	-
[64]	Optimized RF	0.856	0.828	-
[17]	Gradient Boosted DT with SMOTE	0.824	0.838	-
[59]	Rotation Forest	0.770	-	-
[59]	Random Subspace DT	0.761	-	-
[17]	Gradient Boosted DT with SMOTETomek	0.835	0.868	-
[55]	Boosted LR Ensemble	0.810	-	-
[65]	Deep LSTM	0.871	-	-
[60]	RF and Feature selection	0.788	0.735	0.764
This study	Proposed Stacked Ensemble	0.914	0.928	0.891

integration of SMOTE-ENN resampling with the proposed stacked ensemble model. This stacked ensemble approach consistently outperformed traditional machine learning models and other deep learning configurations, particularly in key metrics like accuracy, recall, and specificity, which are crucial for reliable credit risk assessment. The enhancement in these metrics highlights the effectiveness of using advanced resampling techniques to address class imbalance, a pervasive issue in financial datasets.

The stacked ensemble model's superior performance, especially in predicting credit risk, demonstrates its ability to leverage the strengths of multiple base learners, such as CNNs, random forests, and logistic regression, to capture complex patterns in the data. This capability is essential in financial applications, where nuanced patterns and interactions often determine creditworthiness. The significant gains in accuracy and the balanced improvement in sensitivity and specificity emphasize that ensemble methods, combined with robust data preprocessing like SMOTE-ENN, can provide a more reliable and holistic approach to credit risk prediction. This not only enhances predictive accuracy but also contributes to the model's robustness and generalizability across different financial scenarios.

The proposed method integrates SMOTE-ENN to balance the dataset and also makes use of a stacked ensemble model to enhance the prediction of credit risk. Also, the use of CNN for the tabular data feature extraction helped to strengthen predictive accuracy. Compared to existing works, this approach effectively handled class imbalance and leveraged deep learning ensemble techniques, which help to identify high-risk borrowers while minimizing false positives.

VI. CONCLUSION

This study has demonstrated that integrating SMOTE-ENN resampling with a stacked ensemble model significantly enhances the accuracy, sensitivity, and specificity of credit risk predictions on imbalanced datasets. Using the Australian and German credit datasets, the proposed stacked ensemble model outperformed both traditional ML approaches and other DL configurations. This approach not only highlights the effectiveness of combining advanced resampling techniques with sophisticated model architectures but also shows the superior performance achievable in credit risk assessment. The superior results achieved by the stacked ensemble model suggest that such methodologies can better handle the complexities of credit risk prediction. The proposed method has potential applications in large-scale financial systems, including fraud detection, credit risk assessment, and customer segmentation. Its ability to handle imbalanced data is valuable, but real-world deployment may require optimization techniques like distributed computing and real-time learning. However, while SMOTE-ENN effectively addresses class imbalance by generating synthetic samples and removing ambiguous instances, it has potential limitations that warrant consideration. One major concern is synthetic data overfitting, where the model becomes overly reliant on artificially generated samples, leading to poor generalization on real-world data. Additionally, noise introduction from ENN may inadvertently remove informative minority class samples, reducing model robustness. To mitigate these issues, further experiments could compare SMOTE-ENN with alternative resampling techniques such as ADASYN or Borderline-SMOTE. Existing studies also highlight that excessive oversampling can distort feature distributions, impacting model interpretability and performance. Also, future research should focus on incorporating additional data sources and experimenting with even more advanced DL architectures to further refine predictive performance and model robustness. Furthermore, future work should explore ensemble learning, explainable AI for transparency, and benchmarking on real-time financial data to enhance scalability and practical viability.

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